

Ritik Rupam Nanda

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EDUCATION

Centurion University of Technology and Management (CUTM)

Sep 2023 – Sep 2027

Bachelor of Technology in Computer Science

CGPA: 8.44

- Coursework: Machine Learning, Data Visualization & Analysis using Python, Data Structures, Algorithms, DBMS, Web Development

EXPERIENCE

Prodigy InfoTech

May 2025 – Jul 2025

Intern

Mumbai

- Supported staff in daily tasks, reducing workload and allowing greater focus on high-priority assignments.
- Applied learned concepts directly in real-world industry scenarios.

PROJECTS

Banking Application | *Java, Spring Boot, Spring Security, JWT, React.js, MySQL*

- * Developed a full-stack banking application with secure user authentication and role-based access control using JWT and Spring Security.
- * Implemented core banking features including account management, balance inquiry, transaction history, deposits, and fund transfers.
- * Built a responsive React frontend integrated with Spring Boot REST APIs and MySQL database for seamless user experience.

Predictive Detection of DNS Tunneling | *Python, Scikit-learn, XGBoost, TensorFlow, TPOT*

- * Designed an anomaly detection system using entropy and traffic-based features to identify DNS tunneling attacks.
- * Benchmarked SVM, Random Forest, XGBoost, LSTM, and GRU models achieving 99%+ accuracy with automated tuning via TPOT and RandomizedSearchCV.

DeepLens: Interpretable Deepfake Detection | *Python, TensorFlow, Xception, Grad-CAM++*

- * Developed a deepfake detection system using Xception for classification and manipulation-type identification.
- * Achieved 98% accuracy and integrated Grad-CAM++ for interpretable visualization of forged facial areas.

Disaster Damage Assessment using Satellite Imagery | *Python, TensorFlow, Keras, Deep Learning*

- * Built a deep learning pipeline using U-Net and VGG16 + U-Net for pixel-wise segmentation of disaster-affected buildings on the xBD dataset.
- * Optimized with data augmentation and class balancing; validated using F1-Score, IoU, and Dice coefficient.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript

Frontend: React.js, HTML, CSS

Backend: Express.js

Databases: MySQL, PostgreSQL

AI/ML: TensorFlow, Scikit-learn, XGBoost

Tools: Git, VS Code, Jupyter Notebook

Soft Skills: Team Collaboration, Problem Solving, Design Thinking, Visual Design

Languages Known: Odia, English, Hindi

CERTIFICATIONS

- * Kaggle Certified Machine Learning
- * Deloitte Australia Data Analytics Virtual Internship – Forage (July 2025)